# Project Design Document

## 1. Design Intent

- *What experience are you trying to create?- We are creating a fun game that references meme culture and also tests a user's reflexes.*
- *Who is it for?- It's for all age groups, the interface is simple and easy to understand. It's a time killing kind of game*
- *If solving a problem, what problem does it solve?- It does not solve a problem, but it could help with better reflexes.*

## 2. Inspiration & References

- *Existing products / ideas*

*Reflex games, Rave visualizers*

- *Images / links*
- *What did you learn from them?*

*We learnt about the appeal of certain visuals and experiential design in arcade style games.*

## 3. User Journey

*Step-by-step experience of the user:*

1. *User starts by connecting MIT phone App to ESP32 bluetooth.*
2. *User interacts by pressing play on the app and starts pressing the button on the physical product. The button press makes the cat attached to the motor spin.*
3. *System responds by starting an audio on the app and a simultaneous timer in the code that measures the accuracy based on the time window of the physical button press. The user is trying to sync up cat spin to the song.*
4. *In the end, the user gets a score, measuring their accuracy throughout the song.*

The user begins with interactions with the app. Once the user connects the esp32 to their phone via

## 4. Success Criteria

*Define measurable success:*

- *Functional- the motor spins at the same time a click is registered in the code*
- *User experience- The user is able to understand the time window to click if they are familiar with the meme*
- *Reliability- The system functions the same way every time, and returns a score accurate to the user's input.*

## 5. System Overview

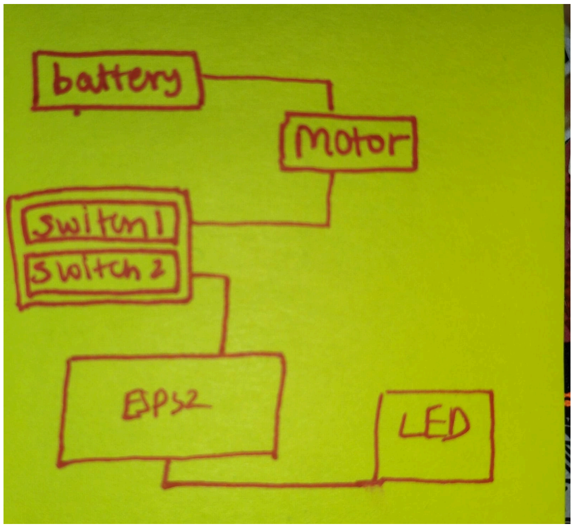
- *High-level explanation of system*

*The system is a 2 circuit contraption meant to simulate the Internet trend of the spinning OIIA cat. The system is considerably inclined to defining user experience, therefore embodying simpler mechanisms and external structure. The centre of this contraption is the button, innovatively allowing the user to engage both circuits at once, as required for planned operation. It "connects" 2 circuits together, one for generating tangible input and the other computational input. This technology facilitates the simultaneous completion of both circuits. The app controls the timer of the entire system, ensuring that every component runs as per accurate duration, the esp32 performs calculations with respect to the accuracy of the user input and the strength of the LED.*

- *What are the main parts?*

The main parts of this system are the motor and the 2switch button. The motor is the only modifiable factor on the user's end which aids them in generating an input. The 2switch button allows the connection of 2 circuits together and delivers the user's input to the system, which is not possible through a single circuit.

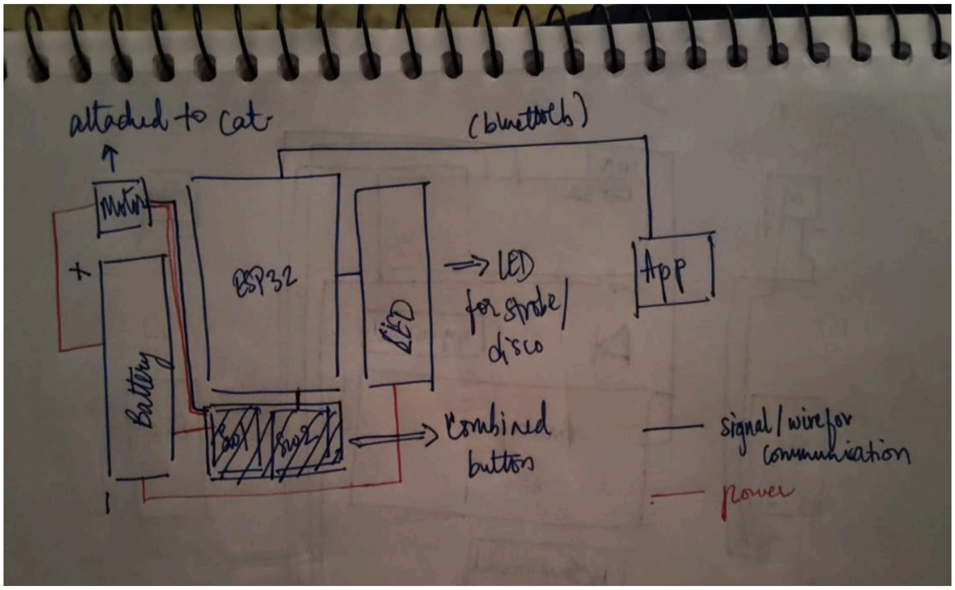
## 6. High-Level Block Diagram




## 7. Input / Output Mapping

| Input | Process | Output |
|---|---|---|
| App button pressed | Launching game | Audio starts playing, strobe lights start flashing |
| Physical button pressed | Timing of click Compared to desirable time window, tally added | Motor spins |
| | Timer runs out | Audio stops, score is displayed, lights stop flashing |

## 8. Final System Architecture (Labelled)




# Project Planning & Management Document

## 1. Project Overview

- *Project Title : OIIA*
- *Team Members & Roles*

*Medha:* Fabrication, electronics, media documentation

*Asmita:* Programming, App, documentation

- *Brief Description (2–3 lines)*

## 2. Team Roles & Responsibilities

| Member Name | Role | Responsibilities |
|---|---|---|
| Medha Nair | | |

| | | |
|---|---|---|
| | Fabrication, electronics, media documentation | In charge of structural integrity of motor support, material choice, innovating conjoint button placement to overcome shorting issues |
| Asmita Wankhede | Programming, App, documentation | In charge of coding time windows, strobes, integration of bluetooth app, audio |

### 3. Project Timeline & Milestones

Achieved within 24 hours due to required pivot from drone.

# Test Plan & Results

### 1. Test Plan

| Test | Method | Expected Result |
|---|---|---|
| *Usability* | **Introduce user to game and allow them to operate it individually** | **User can intuitively understand it's a reflex testing game and successfully connect the app, press the button and obtain their score** |
| *Ergonomics + performance* | **Trying different button placements and running the setup** | **An optimum setup will be comfortable enough for the user to reach and press in a time crunch, and both buttons will be pressed simultaneously every time** |

## 2. User Testing

The users were peers (18-20)

Tasks: Operate the button to play the game

## 3. Results

| Test | Outcome | Pass/Fail |
|---|---|---|
| **Usability** | **The user is able to understand the cultural signifiers the setup embodies and operate it themselves without assistance from us.** | **Pass**<br><br>**Many peers were able to place the cat and the song, and understood what to do without us having to explain much.** |
| **Ergonomics + performance** | **The user is able to press both switches with considerable ease. The angle of the button placement ensures that both switches are pressed.** | **pass** |

## 4. Observations

- *User behavior- Most understood that the cat had to spin at a certain interval of the song because of its cultural popularity.*
- *Pain points- Some users did not know when exactly the game started or that the button would spin the cat.*

## 6. Improvements Suggested

- *Based on testing- Make the button more intuitive and indicate start of timer to the user.*

# Learnings & Reflections Document

## 1. Team Reflection

- *How did the team work together?*

*Instead of relying on individual ability to understand concepts together, due to the lack of time we divided tasks based on individual skill wholly, from structure fabrication and circuitry to app and software code*

- *What worked well?*

*We worked efficiently enough, parallely building the system to not lose out on more time. The division of tasks proved helpful because we did not have to rely on the other for guidance or help and get the allocated stuff done.*

- *What could be improved?*

*We should have allocated more time for testing the system because we had not anticipated it to be so nimble and fall apart every time we tried to prepare it for presentation.*

## 2. Technical Reflection

- *Key technical challenges*

*Centreing both software and hardware around intuitivity despite their structural incapacity*

- *What did you learn (hardware/software)?*

*To work with more structurally rigid and durable material. The complete setup kept falling apart completely everytime we made a minor adjustment. Cardboard would not hold the spinning of the motor or the light effectively enough. The motor wires also got quite cramped and tangled. The software is not organised, there were bits of unused code from trial-and-error which kept affecting the working. Maybe breaking it down into functions would've been a better idea*

- *Mistakes & debugging experiences*

*Because of the messiness in the code, debugging it took almost the same time as writing it, mostly because it would show logical errors only. We had to sit and run through the logic itself for some time so that we could decipher what was going wrong, any help from Claude initially only pushed us further back because we couldn’t explain the context to it properly.*

## 3. Design Reflection

· *Did the experience match your intention?*

*Yes, we were able to create a small, effective simulation of the OIIA cat.*

*The playfulness we aimed to create was delivered.*

· *What would you improve?*

*Increase its intuitiveness and rely less on the user's ability to recognise cultural cues.*

· *User feedback insights*

*A lot of comments on individuals being able to recognise the cat and the rave itself on first glance. They found it cute and funny,*

## 4. Personal Reflection

*Each member writes:*

*Medha*

· *What they learned*

*I learnt to satisfice while still delivering quality. In a span of just days, I managed to execute featured project and it increased my confidence in my decision making and my capacity for work in this sphere.*

· *What they struggled with*

*While reflex games are abundant, it was challenging to find a way to piece together drone parts to give this result. The battery was certainly hard to connect to the motor (which did not have a gpio wire option) without frying the circuit.*

· *What they are proud of*

*I am proud of making a last minute innovation to fix the circuitry and make the spinning register at the same time as the time window in the code even when both of them were different circuits.*

*Asmita*

- *What they learned*

*I think the biggest lesson we learnt here was to have confidence in our own abilities, and that we did not need an extremely complex setup for delivering a good experience,*

- *What they struggled with*

*Code-wise, it required some serious debugging because I hadn't paid enough attention to the nuances of the logic - it became quite cluttered and calculations started almost immediately after the game started. With less time for testing, we also had to cut corners in our logic.*

- *What they are proud of*

*I'm really happy that we were able to deliver something made of our own ability despite encountering many failures with respect to the drone. We pulled through and didn't give up. I also liked that our idea was silly and playful. We hadn't looked at this like a serious academic project but focused on recreating a playful, funny experience that gave our project its own charm.*

Other notes

Why OIIA? : after the drone failed, we needed to create something that demonstrated what we had learnt in ODT sufficiently enough.

Fabrication: the stage itself is crafted out of cardboard and craftpaper, with a slot for the motor to hold the cat.

For its size, we wanted to make it as immersive as we could, therefore using the strobe light sequence, mimicking the actual OIIA rave.

We couldn't connect the motor and battery to the esp32 and risk damaging it, but we still needed a way to let the esp32 know when the motor was spinning in order to record it for the accuracy calculation. We decided to use two buttons as the "response switch" : one would be connected to the motor and battery circuit, and the other be the input for the esp32.